BOUNDED SYSTEMS OF MAPS THAT ARE PRO-ISOMORPHISMS ON COHOMOLOGY ARE PRO-ISOMORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12785-12807, following the statement at 12766-12783.

Editorial note. No omissions: the pro-zero verification is carried out explicitly with the two choices of indices.

Allowed standard prerequisites (editorial).

- Pro-isomorphism criterion via cones. lemma-pro-isomorphism (same chapter, cited at ~12791). Reduces to showing the cone system (C_n) is pro-zero.
- Pro-zero via cohomology. lemma-essentially-constant-cohomology (same chapter, cited at ~12793). Reduces pro-zero of (C_n) to pro-zero of each H^p(C_n).

Source reference. Original lemma 0G3B: sources/stacks-derived/pinned-original-source.tex, source lines 12707–12707. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G3C: sources/stacks-derived/pinned-original-source.tex, source lines 12738–12738. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category.*

$$A_n \to B_n$$

be an inverse system of maps of $D(\mathcal{A})$. Assume

1. *there exist integers $a \leq b$ such that $H^i(A_n) = 0$ and $H^i(B_n) = 0$ for $i \notin [a, b]$, and*
2. *the inverse system of maps $H^i(A_n) \to H^i(B_n)$ of $\mathcal{A}$ define an isomorphism of pro-objects of $\mathcal{A}$ for all $i \in \mathbf{Z}$.*

Then the maps $A_n \to B_n$ determine a pro-isomorphism between the pro-object (A_n) and the pro-object (B_n).

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Proof. We can inductively extend the maps $A_n \to B_n$ to an inverse system of distinguished triangles $A_n \to B_n \to C_n \to A_n[1]$ by axiom TR3. By Lemma 0G3C it suffices to prove that C_n is pro-zero. By Lemma 0G3B it suffices to show that $H^p(C_n)$ is pro-zero for each p. This follows from assumption (2) and the long exact sequences

$$H^p(A_n) \xrightarrow{\alpha_n} H^p(B_n) \xrightarrow{\beta_n} H^p(C_n) \xrightarrow{\delta_n} H^{p+1}(A_n) \xrightarrow{\epsilon_n} H^{p+1}(B_n)$$

Namely, for every n we can find an $m > n$ such that $\mathrm{Im}(\beta_m)$ maps to zero in $H^p(C_n)$ because we may choose m such that $H^p(B_m) \to H^p(B_n)$ factors through $\alpha_n : H^p(A_n) \to H^p(B_n)$. For a similar reason we may then choose $k > m$ such that $\mathrm{Im}(\delta_k)$ maps to zero in $H^{p+1}(A_m)$. Then $H^p(C_k) \to H^p(C_n)$ is zero because $H^p(C_k) \to H^p(C_m)$ maps into $\mathrm{Ker}(\delta_m)$ and $H^p(C_m) \to H^p(C_n)$ annihilates $\mathrm{Ker}(\delta_m) = \mathrm{Im}(\beta_m)$. $\square$